CST2111: DATA VISUALIZATION PROJECT

Final Project

Student guidelines | Fall 2026

SIJU PHILIP

PROCESS

PHASES

EVALUATION CRITERIA

DELIVERABLES

Process



Group formation (Team size 3-4)



Topic selection – refer to ‘Topic to choose from’ slide



Weekly update through presentation during the lecture – max 5 mins (This is divided in Phase 1, Phase 2, and Phase 3. every group member must present his/her own work/work in progress)



If any team member is unable to present, then remaining team members should present their own work. Team member who is absent should send me the update in an email before 3 hours of the lecture with an explanation of his/her absenteeism.



Final presentation in lecture (each member should present his/her own work) – 20 mins



Final Presentation slot – during lecture on December 03, 2026



Project submission – refer ‘Deliverables’ slide

Project Roadmap & Due Dates

Milestone	Due date	Focus	Expected deliverable
Phase 1	October 16, 2026	Problem definition, introduction, objectives	5-minute presentation + two-page draft report
Phase 2	November 6, 2026	Data sources, collection, cleaning, preparation	5-minute presentation + two-page draft report
Phase 3	November 20, 2026	Data modelling, exploration, analysis, visualization	5-minute presentation + two-page draft report
Research paper	December 4, 2026	Technical paper using the required IEEE format	Complete paper with citations and references
Final submission	December 9, 2026	Report, dashboard/workbook/code, and supporting files	Final report, dashboard/workbook/visualization, code or supporting files, and data documentation

What Each Phase Must Achieve

PHASE 1 | DEFINE

Due: October 16, 2026

- Define the problem and its context
- Establish goals and measurable objectives
- Identify the target audience and expected value
- Assign team roles and propose an approach

Deliverables:

- 5-minute presentation
- two-page draft report

PHASE 2 | PREPARE

Due: November 6, 2026

- Identify and justify data sources
- Explain how data was collected
- Assess data quality and limitations
- Clean, transform, and validate data

Deliverables:

- 5-minute presentation
- two-page draft report

PHASE 3 | ANALYZE & COMMUNICATE

Due: November 20, 2026

- Model relationships between datasets
- Explore and analyze the data
- Create relevant charts and dashboards
- Explain findings and implications

Deliverables:

- 5-minute presentation
- two-page draft report

Phase 1 — Problem Definition & Project Proposal

Due October 16, 2026 | In-class presentation: 5 minutes

- **Project title and team:** Use a clear title and list all team members.
- **Problem statement:** Explain the business or research problem, who is affected, and why it matters.
- **Goals and objectives:** Define what the project aims to achieve and list specific questions it will answer.
- **Project overview:** Summarize the proposed solution, target audience, scope, and expected value.
- **Team roles:** Assign responsibilities for data collection, preparation, analysis, visualization, and documentation.
- **Initial approach:** Identify likely data sources and tools, without claiming work that has not yet been completed.

PRESENTATION | 5 MIN

Suggested flow:

- 60 sec: title and context
- 60 sec: problem and objectives
- 60 sec: proposed approach
- 120 sec: roles and expected outcome

TWO-PAGE MINIMUM DRAFT REPORT

Include the problem statement, objectives, project scope, target audience, preliminary data sources, proposed methods, and team responsibilities.

Phase 2 — Data Sources, Collection & Preparation

Due November 6, 2026 | In-class presentation: 5 minutes

- **Data sources:** Name each dataset/source and provide a URL, database, API, or origin where applicable.
- **Relevance:** Explain how the data supports the project's objectives and what it cannot answer.
- **Collection process:** Describe how the data was downloaded, queried, extracted, or gathered.
- **Data quality:** Inspect missing values, duplicates, inconsistent labels, invalid values, and data types.
- **Cleaning and transformation:** Explain the changes actually performed and why they were necessary.
- **Evidence and documentation:** Record row counts, key fields, important limitations, and data preparation decisions.

PRESENTATION | 5 MIN

Suggested flow:

- 60 sec: title and context
- 60 sec: problem and objectives
- 60 sec: proposed approach
- 120 sec: roles and expected outcome

DELIVERABLES

- 5-minute presentation
- Two-page (min) Phase 2 report
- Prepared dataset or evidence of preparation, if required
- Source documentation

Phase 3 — Data Modelling, Analysis & Visualization

Due November 20, 2026 | In-class presentation: 5 minutes

- **Introduction:** Briefly restate the project purpose, objectives, and analytical questions.
- **Data modelling:** Explain table relationships, keys, data structure, and important calculations.
- **Data exploration and analysis:** Use summaries, descriptive statistics, comparisons, trends, or other suitable methods.
- **Tools and technologies:** Name the tools actually used for modelling, analysis, and visualization.
- **Data visualization:** Show relevant charts/dashboard views with clear titles, labels, filters, and units.
- **Findings and interpretation:** Explain what the results mean and how they address the objectives; mention limitations.

5-MINUTE PRESENTATION

0:00–0:30 Purpose
0:30–1:30 Model and tools
1:30–2:30 Analysis
2:30–4:15 Visualizations and findings
4:15–5:00 Conclusion

PHASE 3 DELIVERABLES

- 5-minute presentation
- Two-page (min) Phase 3 progress report
- Data model or relationship explanation
- Analytical evidence and visualizations

Research Paper Submission — IEEE Format

Due December 4, 2026 | Confirm the required IEEE template and page limit.

- **Title, authors, and abstract:** State the project title, team authors, problem, method, key results, and conclusion.
- **Introduction:** Describe the problem, motivation, objectives, and relevant background.
- **Data and methodology:** Explain sources, preparation, data modelling, analytical methods, and tools.
- **Results and discussion:** Present evidence, figures/tables, interpretation, and how results address the objectives.
- **Conclusion and limitations:** Summarize findings, practical implications, constraints, and future work.
- **References:** Use IEEE numbered citations and a matching numbered reference list.

QUALITY EXPECTATIONS

- Follow the required IEEE template
- Label and refer to every figure/table
- Cite data and external material
- Proofread for clarity and consistency

RESEARCH PAPER

The paper is a research-style document having minimum 6 pages and maximum 12 pages .

Final Report — Recommended Structure

Final submission due December 9, 2026

1. Title Page
Project title, team, course, instructor, institution, date.

2. Table of Contents
Headings and page numbers; list of figures/tables if useful.

3. Introduction & Objectives
Context, problem, scope, goals, objectives, research questions.

4. Data Sources & Methodology
Sources, collection, tools, methods, assumptions.

5. Data Modelling & Preparation
Data relationships, keys, cleaning, transformations, quality checks.

6. Data Exploration & Analysis
Descriptive results, trends, comparisons, analytical findings.

7. Visualization & Dashboard
Charts, dashboard screenshots, filters, labels, interpretation.

8. Findings & Recommendations
Insights, implications, objective coverage, actionable recommendations.

9. Conclusion & Limitations
Summary, constraints, lessons learned, future work.

10. References
All cited sources in the required citation style.

11. Appendices
Supporting charts, data dictionary, technical detail, selected code, if needed.

Final Submission — What to Upload

Due December 9, 2026 | Use this checklist to define the required submission package.

- **Final report:** Complete and proofread report using the required structure.
- **Final visualization:** Dashboard, workbook, or visualization file in the approved format.
- **Supporting project files:** Relevant code, data model, workbook, or other files needed to demonstrate the solution.
- **Data documentation:** Data dictionary, source details, and preparation notes where applicable.
- **References and appendices:** Complete citations and necessary supporting evidence.
- **Team submission:** Use clear filenames and ensure team members and responsibilities are documented.

FINAL QUALITY CHECK

Submit the final report, dashboard/workbook/visualization, code or supporting files, and data documentation.

Team Submission

Only the team lead is required to submit all the documents on behalf of the entire team.

Student Master Checklist

Use this as a final self-review before each submission.

PHASE 1 | OCT 16

- [ ] Problem and objectives are clear
- [ ] Team roles are documented
- [ ] 5-minute presentation is prepared
- [ ] Two-page draft report is complete

PHASE 2 | NOV 6

- [ ] Sources and collection are documented
- [ ] Data quality has been assessed
- [ ] Cleaning and transformations are explained
- [ ] 5-minute presentation + two-page report are ready

PHASE 3 | NOV 20

- [ ] Data model and relationships are explained
- [ ] Analysis answers the project questions
- [ ] Visualizations are labelled and interpreted
- [ ] 5-minute presentation + two-page report are ready

PAPER | DEC 4

- [ ] Required IEEE template is used
- [ ] Methods and results are clearly described
- [ ] Figures/tables and citations are consistent
- [ ] References are complete

FINAL | DEC 9

- [ ] Report includes all required sections
- [ ] Final dashboard/workbook is included
- [ ] Required supporting files are included
- [ ] Files are named clearly and proofread